



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
CH101 - ENGINEERING CHEMISTRY

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. Expression of Schrodinger wave equation : (1) C01 BL1 PO1
(a) $\Psi_n = \sqrt{(2/a)} \sin(n\pi x/a)$ (b) $\Psi_n = \sqrt{(2/a)} \sin(n\pi x)$
(c) $\Psi_n = \sin(n\pi x/a)$ (d) $\Psi_n = \sqrt{a} \sin(n\pi x/a)$
2. The reaction proceeds spontaneously when (1) C02 BL1 PO1
(a) $\Delta G > 0$ (b) $\Delta G < 0$
(c) $\Delta G = 0$ (d) None of these
3. Which of the following compound is the strongest acid? (1) C04 BL1 PO2
(a) Phenol (b) Ethanol
(c) Acetic acid (d) Methanol
4. Which element has the highest electron affinity? (1) C01 BL1 PO1
(a) F (b) Br
(c) Cl (d) I
5. Which property of HDPE makes it suitable for making water pipes? (1) C03 BL1 PO1
(a) Low tensile strength (b) High elasticity
(c) High tensile strength (d) Soft nature
6. Which of the following is a conducting polymer? (1) C04 BL2 PO2
(a) Bakelite (b) Polyaniline

- (c) Polystyrene (d) Polythene
7. Conversion of electrical energy to chemical energy is done by- (1) C02 BL2 PO3
 (a) Electrolytic cell (b) Galvanic cell
 (c) Hydro dynamic cell (d) All of these
8. The Second Law of Thermodynamics is called the Law of— (1) C02 BL2 PO1
 (a) Enthalpy (b) Entropy
 (c) Internal Energy (d) Gibbs Energy
 (e) None of these
9. The basic principle of MRI is related to: (1) C05 BL1 PO1
 (a)) UV-vis spectroscopy (b) NMR spectroscopy
 (c) IR spectroscopy (d) microwave spectroscopy
10. Inversion of configuration is associated with which of the following? (1) C03 BL1 PO1
 (a) E1 reaction (b) SN₂ reaction
 (c) SN₁ reaction (d) free radical halogenations
11. Which one of the following compounds is the most reactive towards SN2 mechanism? (1) C04 BL1 PO1
 (a) n-butyl chloride (b) t-butyl chloride
 (c) 2-chlorobutane (d) 2-methyl-1-chloropropane
12. Major component of CNG is (1) C03 BL1 PO1
 (a) Methane (b) Ethane
 (c) Ethylene (d) Propen

Part B**(Answer any four of the following) 5x4=20**

13. (a) "Finger Print Region " term is associated with which spectroscopy? Define it. (2) C04 BL2 PO2
 (b) If you are given two samples one is an organic acid another is an organic alcohol, how will you distinguish between them with the help of above mentioned spectroscopy. (3) C04 BL3 PO2
14. Propene + HBr in presence of Peroxide = Products? Discuss mechanism. (5) C04 BL4 PO4
 Propene + HBr = Products? Discuss mechanism.

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|-----|--|-----|-----|-----|-----|
| 15. | How we can explain the correct electronic configuration of $_{24}\text{Cr}$ with the help of exchange energy concept? | (5) | C02 | BL3 | PO1 |
| 16. | Write the differences between exothermic reactions and endothermic reactions. Explain using a graph. | (5) | C02 | BL2 | PO1 |
| 17. | Calculate the EMF of the following cell:
$\text{Ni} \mid \text{Ni}^{2+} (1\text{M}) \parallel \text{Pb}^{2+} (1\text{M}) \mid \text{Pb}$ at 25°C
Given $E^\circ \text{Ni}^{2+} \mid \text{Ni} = -0.24\text{V}$, $E^\circ \text{Pb}^{2+} \mid \text{Pb} = -0.13\text{V}$, state also whether the reaction is spontaneous or feasible. | (5) | C02 | BL4 | PO3 |
| 18. | $\text{S}_\text{N}2$ reaction occurs with inversion of configuration and $\text{S}_\text{N}1$ reaction occurs with of configuration. - Explain | (5) | C04 | BL2 | PO2 |

Part C**(Answer any three of the following) 3x10=30**

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|-----|--|---------------------------|---------------------------|---------------------------|---------------------------|
| 19. | A. Write down the monomers of the following polymers with chemical structure.

i) PVC ii) Natural Rubber iii) PVC iv) Buna N
v) Polystyrene

B. Write the Nernst equation and calculate the emf of the following cell at 298 K.

$\text{Cu (s)} \mid \text{Cu}^{2+} (0.130 \text{ M}) \parallel \text{Ag}^+ (1.00\text{M}) \mid \text{Ag (s)}$

Given: $E^\circ \text{Cu}^{2+}/\text{Cu} = 0.34\text{V}$ and $E^\circ \text{Ag}^+/\text{Ag} = +0.80 \text{ V}$. | (10) | C03 | BL2 | PO3 |
| 20. | (a) Write differences between n-type and p-type semiconductors with example.

(b) What are the precursors for the preparation of Nylon 6,6? Write two important applications of Nylon 6,6.

(c) What is octane number? Explain with suitable example. How it can be improved? | (2)

(4)

(4) | C05

C01

C02 | BL1

BL1

BL3 | PO2

PO1

PO1 |
| 21. | (a) Derive the expression for the efficiency of the heat engine with the help of Carnot's cycle.

(b) State and explain Hess's law of constant heat summation

(c) Write one example of extensive property. | (6)

(3)

(1) | C02

C02

C02 | BL4

BL2

BL1 | PO2

PO2

PO2 |

22. A. Write the steps involved in the preparation of Paracetamol. (10) C03 BL4 PO3
State its uses.
B. What happens when HBr is added to But-1-ene in presence of benzoyl peroxide. Explain using a chemical reaction.
C. Comment on the basicity of aniline and methylamine. Explain your answer.
23. (a) What is Absorbance? Provide schematic graphical presentation of UV-Vis and IR spectra. (5) C05 BL1 PO1
- (b) What is Chromophore and Auxochrome? Explain with example (5) C05 BL2 PO1